DEPI

Digital Egypt Pioneers Initiative

UniSwap – Student Marketplace

Project Documentation – Part I

Track: React Front-End Web Developer

Technical Instructor: Mohamed Mahmoud

Project Type: Team Graduation Project

Team Size: 5 Members

Team Leader: Gamila Abdel Khalek

GitHub: https://github.com/SWESiham/UniSwap

Submission Schedule (Part I)

Item	DEPI Industry	DEPI
Project Planning & Management	2 Oct 2026	16 Oct 2026
Literature Review	9 Oct 2026	16 Oct 2026
Requirements Gathering	9 Oct 2026	16 Oct 2026

Contents

- 1. Project Planning & Management – Project Proposal, Project Plan, Task Assignment & Roles, Risk Assessment & Mitigation Plan, KPIs
- 2. Literature Review – Review of Similar Solutions, Feedback & Evaluation, Suggested Improvements, Final Grading Criteria
- 3. Requirements Gathering – Stakeholder Analysis, User Stories & Use Cases, Functional Requirements, Non-functional Requirements
- 4. Repository & Project Information

1. Project Planning & Management

1.1 Project Proposal

UniSwap is a student-focused marketplace that gives university students one place to discover, sell, request, and interact around items within their student community. The platform combines marketplace browsing, search and filtering, favorites, student requests, communication, administration, and AI-assisted listing creation.

Problem Statement

University students often rely on scattered channels to sell or find used books, calculators, electronics, stationery, lab tools, notes, and other student-related items. These channels do not provide a dedicated marketplace experience with structured listings, search and filtering, favorites, requests, moderation, and direct communication. This makes it harder for buyers to find relevant items and for sellers to manage their listings.

Proposed Solution

UniSwap provides a centralized web marketplace tailored to university students. Students can create and browse structured listings, search and filter items, save favorites, publish requests for needed items, and communicate with other users. Administrators can moderate listings, manage users and categories, and handle reports. The Gemini API supports AI-assisted listing analysis and autofill.

Objectives

- Build a centralized marketplace for university students.
- Make discovering relevant listings faster through search, filters, sorting, and pagination.
- Provide a structured way to publish and manage listings.
- Support AI-assisted listing creation using image analysis and structured data extraction.
- Enable favorites, requests, and student-to-student communication.
- Provide administrative moderation and management capabilities.
- Apply authentication, authorization, validation, and secure API practices.

Scope

In scope: authentication and profiles, listing management, AI-assisted listing analysis, marketplace search and filtering, favorites, requests, conversations and messaging, and an admin dashboard (users, listing moderation, reports, categories), with a responsive interface, Arabic/English support, RTL/LTR, and light/dark mode.

Optional (MVP-dependent): Exchange Proposal (propose, accept, decline). It is removed first if the schedule is at risk.

Out of scope: online payments and delivery.

Technology Stack

Area	Technology
Frontend	React + TypeScript
Backend	Node.js + Express (Socket.IO for chat)
Database	MongoDB
AI	Gemini API
Authentication	JWT, bcrypt password hashing
Version Control	Git + GitHub, feature branches + Pull Requests

1.2 Project Plan

The project runs for **9 weeks, from 4 Oct to 4 Dec 2026**. The plan covers the whole project (analysis, design, implementation, testing, and final delivery) and is independent of the individual documentation deadlines. Implementation is finished and deployed by **28 Nov**, ahead of the 30 Nov implementation deadline.

Phases

Phase	Dates	Goal
1. Planning & Setup	4 – 17 Oct (W1–W2)	Idea, scope, repository, branches, task split, data models and API contracts agreed
2. Backend Foundation	18 – 24 Oct (W3)	Authentication, core models, and core APIs working and tested in Postman
3. Core Features	25 Oct – 7 Nov (W4–W5)	Main screens and APIs built for each feature, including AI analysis and chat
4. Integration & Completion	8 – 21 Nov (W6–W7)	Frontend connected to backend, all features complete, feature freeze
5. Testing & Deployment	22 – 28 Nov (W8)	Test cases, bug fixing, deployment, README
6. Final Delivery	29 Nov – 4 Dec (W9)	Final reports, presentation, demo, rehearsal

Gantt Chart

Workstream	W1	W2	W3	W4	W5	W6	W7	W8	W9
Planning, scope, repo setup									
Auth & Profile									
Listings & AI									
Marketplace & Favorites									
Requests, Chat & Exchange									
Admin & Moderation									
Shared UI (Navbar, theme, i18n, RTL)									
Integration between features									
Testing & bug fixing									
Deployment & README									
Final reports & presentation									
Buffer & rehearsal									

W1 = 4–10 Oct, W2 = 11–17 Oct, W3 = 18–24 Oct, W4 = 25–31 Oct, W5 = 1–7 Nov, W6 = 8–14 Nov, W7 = 15–21 Nov, W8 = 22–28 Nov, W9 = 29 Nov–4 Dec.

Milestones

#	Milestone	Date
M1	Repository, branches, and API contracts agreed	17 Oct
M2	Auth + User, Listing, and Category models working	24 Oct
M3	Marketplace browsing, Add Item, and Requests basic flow working	31 Oct
M4	AI analysis (Gemini) and chat working	7 Nov
M5	All five features integrated	14 Nov
M6	Feature freeze (all features complete)	21 Nov
M7	Code freeze and deployed version live	28 Nov
M8	Final presentation, demo, and reports ready	4 Dec

Weekly Plan per Developer

Week	DEV 1 Auth & Profile	DEV 2 Listings & AI	DEV 3 Marketplace	DEV 4 Requests & Chat	DEV 5 Admin
W1 **4–10 Oct**	Team-wide: idea, scope, tech stack, repo, branches	–	–	–	–
W2 **11–17 Oct**	User model, register/login API, JWT	Listing model, Listing CRUD API	Favorite model, GET /listings	Request model and APIs	Category model, admin layout
W3 **18–24 Oct**	Auth middleware, role-based authorization, Login/Register UI	Image upload, Add Item UI	Search, filter, sort, pagination API	Conversation and Message models, Socket.IO setup	Admin users APIs, admin stats API
W4 **25–31 Oct**	Profile APIs, Profile/Edit Profile/Settings UI	Product Details, Edit Item, Mark as Sold, AI analyze API	Marketplace UI, grid/list, categories	Requests Board, Create/Edit/Delete Request UI	Listing moderation APIs, Report model
W5 **1–7 Nov**	User Dashboard, My Listings, My Requests pages	AI Analysis UI, Autofill, Price Suggestion	Filter UI, Favorites	Chat UI, product attached to chat	Dashboard statistics and charts
W6 **8–14 Nov**	Forgot Password, Admin Login, Navbar, Dark/Light mode	Review Before Publish, AI fallback, error states	Sorting, Pagination UI, Loading/Empty states	I Have This flow, Exchange Proposal (optional)	Users management UI, Block/Unblock, Approve/Reject UI
W7 **15–21 Nov**	My Favorites, Sold Items, security review	Arabic/English + RTL, UI polish	Responsive design, Landing page, Footer	Accept/Decline Exchange (or cut), Contact/About	Reports, Categories Management, Admin Settings, 404 page
W8 **22–28 Nov**	Test cases (Auth), bug fixes, deployment, README	Test cases (Listings/AI), bug fixes	Test cases (Marketplace), bug fixes	Test cases (Requests/Chat), bug fixes	Test cases (Admin), merged Test Plan, API documentation
W9 **29 Nov–4 Dec**	Technical Documentation	Video Demo (optional)	User Manual	Project Infographic	Presentation (PPT/PDF)

Feature freeze is 21 Nov. Everyone joins the rehearsal on 2–3 Dec, and 4 Dec is final delivery.

Deliverables

Deliverable	Owner	Due
GitHub repository with branches and PR history	All (setup: DEV 1)	Continuous
Backend APIs for all five features	Each developer	21 Nov
Frontend for all screens	Each developer	21 Nov
Gemini AI integration with fallback	DEV 2	14 Nov
Admin dashboard	DEV 5	21 Nov
Test cases, test plan, bug reports	Each developer, merged by DEV 5	28 Nov
API documentation	Each developer, merged by DEV 5	28 Nov
Deployed web app and README	DEV 1	28 Nov
Technical Documentation	DEV 1	3 Dec
User Manual	DEV 3	3 Dec

Deliverable	Owner	Due
Project Infographic	DEV 4	3 Dec
Presentation (PPT/PDF)	DEV 5	3 Dec
Video demo (optional)	DEV 2	3 Dec

Resource Allocation

Resource	Allocation
Team	5 developers; each owns one feature across frontend, backend, database, API, and integration
Version control	GitHub with feature branches. No direct push to main; every merge goes through a reviewed Pull Request
External services	Gemini API (AI analysis) and MongoDB database
Meetings	Weekly planning (Sunday), mid-week sync on blockers, Thursday demo and merge of reviewed PRs
Review process	At least one other developer reviews each Pull Request before merging

Dependencies and Scope Control

- The User model and JWT middleware (DEV 1) must be ready by 24 Oct because every protected API depends on them.
- The Listing model and its status field (pending, approved, rejected, sold) are agreed between DEV 2 and DEV 5 in W2.
- The Category model (DEV 5) is ready in W2 because marketplace filters use it.
- The Gemini API key and quota are confirmed by DEV 2 in W2 so AI work is not blocked.
- If time is short, features are cut in this order: Exchange Proposal, Admin Settings and chart polish, Sold Items and Settings pages, video demo.

1.3 Task Assignment & Roles

Team Leader: Gamila Abdel Khalek. Each developer owns a complete feature from frontend to database and API integration, so nobody is assigned to frontend or backend only.

DEV	Member	Area	Main Responsibilities
DEV 1	Basel Abdo	Authentication & User Profile	Register, login, forgot password, admin login, JWT, bcrypt, auth middleware, role-based authorization, user and profile APIs, user dashboard pages, deployment and README
DEV 2	Siham Sayed	Listings & AI	Listing CRUD and ownership, image upload, Gemini image analysis, AI autofill and price suggestion, review before publish, mark as sold, AI failure fallback
DEV 3	Mariem Mohamed	Marketplace, Search & Favorites	Browse, search, filters (price, condition, faculty, department), sorting, pagination, favorites, loading and empty states

DEV	Member	Area	Main Responsibilities
DEV 4	Gamila Abdel Khalek	Requests & Chat	Requests board and request management, "I Have This", conversations, messages (Socket.IO), product attached to chat, exchange proposal (optional)
DEV 5	Basem Obeid	Admin Dashboard & Moderation	Statistics and charts, user management and block/unblock, listing approve/reject, reports, category management, merged test plan and API documentation

Shared Work

Shared Item	Owner
Navbar, dark/light mode	DEV 1
Arabic/English, RTL/LTR	DEV 2
Landing page, footer, responsive design	DEV 3
Contact / About	DEV 4
404 page, common error handling	DEV 5

Development Approach and Git Workflow

The team uses a feature-based workflow with separate feature branches, meaningful commits, Pull Requests, code review, testing, and controlled merging into main.

- main – stable integrated branch.
- feature/auth-profile – DEV 1.
- feature/listings-ai – DEV 2.
- feature/marketplace – DEV 3.
- feature/communication – DEV 4.
- feature/admin – DEV 5.
- Flow: develop, self-test, push branch, Pull Request, review, fix comments, merge.

1.4 Risk Assessment & Mitigation Plan

Risk	Impact	Likelihood	Mitigation
Integration conflicts between features	High	Medium	Agree on API contracts early; use feature branches and Pull Requests; integration weeks W4–W7
Limited experience with some backend concepts	Medium	Medium	Work incrementally, review code together, test each feature before integration
AI analysis failure or inconsistent output (Gemini)	Medium	Medium	Validate AI output and provide a manual fallback for listing creation; confirm API key and quota in W2
Scope becoming too large	High	Medium	Prioritize the MVP; cut Exchange Proposal first if needed (see Scope Control)
Merge or regression issues	Medium	Medium	Require self-testing and code review before merging

Risk	Impact	Likelihood	Mitigation
Time pressure near deadlines	High	High	Track milestones weekly; feature freeze 21 Nov; deployment by 28 Nov; buffer week W9
A team member unavailable or behind schedule	High	Low	Raise delays immediately; rebalance tasks or reduce scope; shared ownership of reviews
Deployment or hosting problems	Medium	Medium	Start deployment preparation in W7; keep a local run guide in the README

1.5 Key Performance Indicators (KPIs)

KPI	Target	How It Is Measured
Feature completion	100% of MVP features complete by the 21 Nov freeze	Checklist per feature, reviewed weekly
API response time	Under 2 seconds for marketplace browsing and search	Postman / browser network tab
Search and filter accuracy	Results match the selected filters and show only active, approved listings	Test cases
Authentication and authorization	Protected routes reject unauthenticated users; admin routes reject non-admins	Test cases
AI listing autofill	AI result returned or fallback shown in 100% of attempts	Test cases with valid and failing inputs
Test coverage	Test cases written and run for every feature; critical bugs fixed before release	Test plan and bug report
System availability	Deployed app accessible during the evaluation period	Manual checks, deployment link
Repository and documentation	All source code and documents in GitHub; every merge via reviewed PR	Repository history
User adoption (pilot)	Pilot test with classmates; positive feedback on core flows	Short feedback form after deployment

2. Literature Review

The team reviewed the concept of online marketplaces and student-oriented buying and selling platforms to identify useful patterns and gaps. Existing services show the value of structured listings, search, categories, filtering, user accounts, and seller-buyer interaction. UniSwap is designed specifically around university students and their context rather than as a general-purpose marketplace.

2.1 Review of Existing Solution Patterns

Solution Type	Strengths	Gaps for Students
General marketplaces	Structured listings, categories, search, filters, seller information, communication	Not tailored to faculty or department; no student requests board; mixed audience
Social media and messaging groups	Easy to start, large student communities	Unstructured posts, weak search, no moderation tools, items get buried

Solution Type	Strengths	Gaps for Students
UniSwap (proposed)	Student-focused marketplace with faculty/department filters, requests, favorites, chat, admin moderation, and AI-assisted listing	New platform; needs users and trust-building over time

Moderation and account management are also important in all marketplaces for maintaining trust and controlling inappropriate listings.

2.2 Identified Opportunity

UniSwap focuses the marketplace on university students and adds student-specific context: faculty and department filters, requests for needed items, favorites, and communication. It also introduces AI-assisted listing creation to reduce manual data entry.

2.3 Feedback & Evaluation

This section records the lecturer's assessment of the project. It is completed after the project is presented to the Technical Instructor, Mohamed Mahmoud.

Area	Lecturer Feedback	Team Response
Project idea and scope		
Planning and documentation		
Requirements		
Overall evaluation		

2.4 Suggested Improvements

Areas where the project can be enhanced, based on the team's own review (to be extended with the lecturer's suggestions):

- Add the Exchange Proposal feature (propose, accept, decline) if time allows.
- Add ratings and reviews for sellers to build trust.
- Add email or in-app notifications for messages, requests, and listing approval.
- Improve AI features, for example better price suggestions and duplicate listing detection.
- Add a mobile application or a Progressive Web App version.
- Integrate university login or student email verification.

2.5 Final Grading Criteria

The official breakdown of marks across documentation, implementation, testing, and presentation is provided by the Technical Instructor. The table is completed once the grading scheme is confirmed.

Component	Marks	Related Project Deliverables
Documentation		Planning, literature review, requirements, system analysis and design, reports
Implementation		Source code, GitHub repository, deployed application, README
Testing		Test cases and test plan, automated tests, bug reports
Presentation		Presentation (PPT/PDF), infographic, optional video demo
Total		

3. Requirements Gathering

3.1 Stakeholder Analysis

Stakeholder	Needs	Influence / Interest
Students / Buyers	Find relevant items quickly; trust listings; contact sellers; request items they need	High interest, medium influence
Students / Sellers	Publish listings easily (with AI help); manage and mark items as sold; reach interested buyers	High interest, medium influence
Platform Administrators	Moderate listings; manage users, categories, and reports; view platform statistics	High interest, high influence
Technical Instructor	Clear documentation, working implementation, good engineering practices	High interest, high influence
Project Team	Clear task ownership, manageable scope, stable integration	High interest, high influence

3.2 User Stories & Use Cases

User Stories

- As a student, I want to register and log in so that I can securely use my account.
- As a student, I want to browse active listings so that I can discover available items.
- As a student, I want to search and filter listings so that I can quickly find relevant products.
- As a student, I want to save listings as favorites so that I can return to them later.
- As a student, I want to create and manage my listings so that I can sell items.
- As a student, I want AI assistance when creating a listing so that product information can be filled more efficiently.
- As a student, I want to mark my item as sold so that other students know it is no longer available.
- As a student, I want to create a request for an item I need so that other students can respond.
- As a student, I want to communicate with another student so that I can discuss a listing or request.

- As an administrator, I want to moderate listings so that inappropriate or invalid content can be controlled.
- As an administrator, I want to manage users and categories so that the platform remains organized.
- As an administrator, I want to review reports so that I can act on reported content.

Use Cases

Actor	Use Case	Description
Student	Register / Login	Create an account and authenticate
Student	Browse Listings	View available marketplace items
Student	Search & Filter	Find listings using search and marketplace filters
Student	Manage Favorites	Save or remove listings from favorites
Student	Manage Listings	Create, edit, view, and delete owned listings; mark as sold
Student	Use AI Listing Assistant	Upload an image, receive AI analysis, review and publish
Student	Create Request	Publish a request for an item needed
Student	Chat	Communicate with other students
Student	Propose Exchange (optional)	Propose an exchange; the other student accepts or declines
Admin	Manage Users	Search, filter, block, and unblock users
Admin	Moderate Listings	Approve or reject listings
Admin	Manage Reports	Review and manage reported content
Admin	Manage Categories	Create, update, and delete categories

Sample Use Case Description: Create Listing with AI Assistance

Field	Description
Actor	Student (authenticated)
Precondition	The student is logged in
Main flow	1. Student opens Add Item and uploads an image. 2. The system sends the image for AI analysis. 3. The system returns structured data and fills the form. 4. The student reviews and edits the data. 5. The student publishes; the listing awaits approval.
Alternative flow	If AI analysis fails, the system shows a message and the student fills the form manually
Postcondition	The listing is saved and visible after admin approval

3.3 Functional Requirements

FR-01: The system shall allow students to register and log in.

FR-02: The system shall authenticate users using JWT-based authentication.

FR-03: The system shall support role-based authorization for administrative operations.

FR-04: The system shall allow users to create, edit, view, and delete their listings according to ownership rules.

FR-05: The system shall support image upload for listings.

FR-06: The system shall provide AI-assisted listing analysis and structured data extraction, with a manual fallback if analysis fails.

FR-07: The system shall allow users to browse active and approved listings.

FR-08: The system shall provide search by listing information.

FR-09: The system shall provide filtering by category, price range, condition, faculty, and department.

FR-10: The system shall support sorting and pagination of marketplace results.

FR-11: The system shall allow authenticated users to add and remove favorites.

FR-12: The system shall allow students to create, view, edit, and delete requests according to permissions.

FR-13: The system shall support conversations and messages between users.

FR-14: The system shall provide administrators with user management, including block and unblock.

FR-15: The system shall support listing approval/rejection, reports, and category management.

FR-16: The system shall allow a seller to mark a listing as sold.

FR-17: The system shall display admin statistics on the admin dashboard.

FR-18 (optional): The system shall support exchange proposals that can be accepted or declined.

3.4 Non-functional Requirements

Category	Requirement
Security	Authentication, role-based authorization, bcrypt password hashing, input validation, and protected operations
Performance	Efficient search, filtering, pagination, and API responses; marketplace pages respond within about 2 seconds
Usability	Clear navigation, loading states, empty states, and understandable success/error messages
Reliability	Graceful error handling and fallback behavior when the external AI service fails
Maintainability	Modular code, reusable components, meaningful naming, and clear project structure
Scalability	Modular architecture that allows adding marketplace features later
Compatibility	Responsive web interface for common desktop and mobile screen sizes
Localization	Arabic/English support with RTL/LTR interface direction, and light/dark mode

4. Repository & Project Information

GitHub Repository: https://github.com/SWESiham/UniSwap

The repository contains the project source code and project documentation. The team maintains a feature-branch workflow and uses Pull Requests for controlled integration.

Next Documentation Phase

The next phase expands the requirements into System Analysis & Design deliverables: software architecture, ER diagram and schema, use case diagram, DFD, sequence, activity, state and class diagrams, UI/UX mockups, component and deployment views, API documentation, and the testing plan.